ECN 712 Fall 2025
EE Schlee
Problem Set VI, Due Thursday, 23 October

1. Let $\mathbb{X} = \{x_1, ..., x_n\}$ be a finite set of outcomes, $n \geq 3$, $\mathcal{L}$ the set of probablity distributions over $\mathbb{X}$ and $\succsim \subset \mathcal{L} \times \mathcal{L}$ a complete and transititve binary relation. Suppose that there is a an expected utility representation of $\succsim$: there is a real-valued function u on $\mathbb{X}$ such that for p, q in $\mathcal{L}$, $q \succsim p$ if and only if $\sum_i u(x_i)q_i \geq \sum_i u(x_i)p_i$. Suppose too that not all elements of $\mathcal{L}$ are indifferent.

   (a) Prove that $\succsim$ is (mixture) continuous and satisfies independence.

   (b) Prove that u is unique up to a (strictly) increasing affine transformation: a real-valued function v on $\mathbb{X}$ satisfies "for p, q in $\mathcal{L}$, $q \succsim p$ if and only if $\sum_i v(x_i)q_i \geq \sum_i v(x_i)p_i$" if and only if "$v = a + bu$ for a real number a and a positive real number b." (The "if" part is easy, the "only if" part is harder, but if you get stuck look at the closely related Proposition 6.B.2 in MWG, though there is a *much* shorter argument than theirs: I suggest you start with utilities of the best and worst outcomes and use these to define a and b.)

2. Let $\mathbb{X} = \{x_1, ..., x_n\}$ be a set of outcomes ($n \geq 3$), with x_1 the outcome "death." Define $\succsim$ as follows: for any $q, p \in \mathcal{L}$, if $q_1 > p_1$, then $p \succ q$; if $q_1 = p_1$, then there is a real-valued function u on $\{x_2, ..., x_n\}$ such that $p \succsim q$ precisely when

$$\sum_{i=2}^{n} u(x_i)p_i \geq \sum_{i=2}^{n} u(x_i)q_i.$$

   where u is a *nonconstant* real-valued function on $\{x_2, ..., x_n\}$.

   (a) Does $\succsim$ satisfy the independence axiom? Prove it or explain why not.

   (b) Suppose that no two degenerate elements of $\mathcal{L}$ (that is, the unit vectors, e_i in my notation) are indifferent to one another. Is there an expected utility representation of $\succsim$?

   (c) How do you reconcile your answers to (a) and (b)? How if at all would your answer change to (b) change if $n = 2$?

3. Prove *directly* that the "modal" choices in the Allais paradox ($p \succ q$ but $s \succ r$ in my class notation) violate the independence axiom. (If you get stuck you might start with the easier fact that the modal choices are inconsistent with an *expected utility representation* of those preferences.)

4. (a) MWG, Exercise 6.B.6. (The simpler your example, the better; and yes I know you've proved the first sentence before; do it again, and strive to improve upon your last argument.)

   (b) Let $a^*(\cdot)$ be the solution correspondence for the problem at the bottom of page 182. Give a sufficient and necessary condition on $a^*(\cdot)$ for U to be affine. Give a sufficient and necessary condition on $a^*(\cdot)$ for U to be strictly convex. Interpret your conditions. (Hint: Example 6.B.4.)

5. (a) MWG, Exercise 6.C.1, with this, more natural notation: let w be initial wealth, p the loss chance, I the coverage level, and r the per-unit price of coverage. In this notation prove this: if $r > p$, then the consumer demands *less than* full coverage.

   (b) Translate this insurance problem into the notation of the portfolio problem we discussed together. That is, write final wealth in state s as $w' + \alpha x_s$ for $s = 1, 2$, where $s = 1$ is the loss state and $s = 2$ is the no-loss state. (Hint: don't set $w' = w$.) After this translation you should see that MWG 6.C.1 is a *special* case of the result we proved together, that generically, anyone with risk-averse, monotone EU preferences will take some fraction of a positive-mean bet.

6. Let u be a continuous and strictly increasing (vN-M) utility function over consumption plans in R^L_+, $L \geq 2$, and $v(p, w)$ its associated indirect utility function.

   (a) Prove that if u is concave then v is concave in w for each $p \gg 0$: if a consumer is averse to consumption risk, then the consumer is averse to wealth risk. (Risk aversion for outcomes in a convex set X of a Euclidean space just means that the consumer always weakly prefers a riskless $x \in X$ to a risk with the same mean.)

   (b) Suppose now that prices are random, but wealth is not. *Evaluate*: "The consumer cannot be *strictly* averse to price risk." ("Strict" risk aversion means that the consumer *strictly* prefers a sure thing to any *distinct* risk with the same mean. The conclusion follows as an almost-immediate corollary to something we proved together in the first part of the course; your answer should just be a line or two if you appeal to what we proved together.)

   (c) Suppose that the consumer is neutral to wealth risk for every fixed $p \gg 0$. What form must the indirect utility function take? What does this form imply about this consumer's Engel curves?

7. A consumer has initial wealth of w and divides his wealth between a safe asset with a rate of return of 0, and a risky asset with a rate of return of $x + t$. The number t is a constant and the distribution of x is given by a cumulative distribution function (c.d.f.) F with bounded support satisfying $\int x dF(x) = 0$ and $0 < F(0) < 1$. The consumer has expected utility preferences on $\mathcal{L}$ with a C^2 vN-M utility u satisfying $u' > 0$ and $u'' < 0$.[1] Let $\alpha^*(t)$ denote the consumer's expenditure on the risky asset as a function of the constant t.

   (a) Confirm that $\alpha^*(0) = 0$—the simpler your argument the better—and $\alpha^*(t) > 0$ for $t > 0$.

   (b) Calculate $\alpha^{*\prime}(0)$—hint: implicit function theorem—and confirm that it is positive. Show that its magnitude depends on the value of the Arrow-Pratt risk aversion measure at the point w; and the *variance* of the random return, x. Explain the relationship intuitively.

[1] $\mathcal{L}$ *here* is the set of cumulative distribution functions with bounded support in $\mathbb{R}_+$.